GlucoTwin

Personalized glucose forecasting between clinic visits

Digital Twin Challenge 2026 · Happiest Health

Team Kryptonite — Avinash Gehi (team leader) · Hardik Hinduja

College / incubator: to be confirmed by the team

Student research prototype — not a medical device

The problem: what happens between appointments

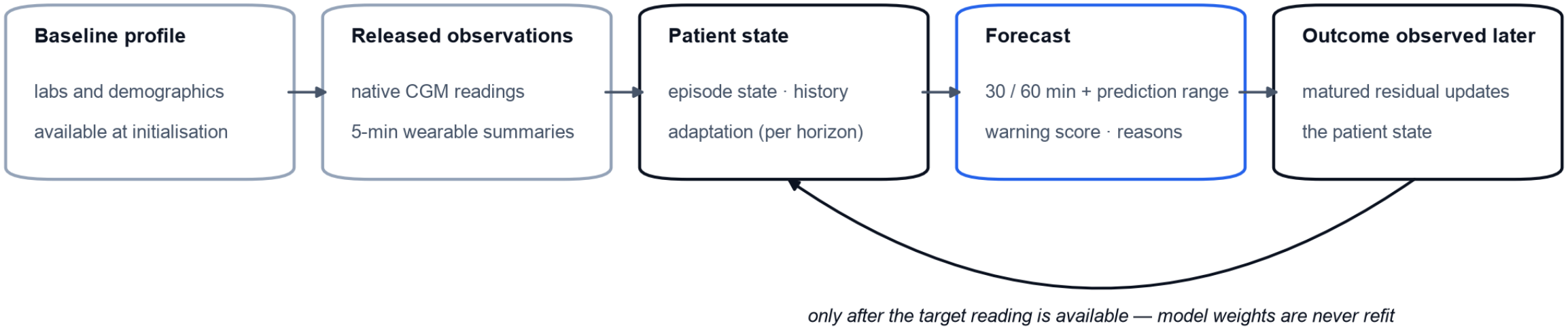
- HbA1c summarises ~3 months of glucose; it cannot show short high-glucose excursions (NIDDK).
- WHO identifies continuity of care and diabetes monitoring as ongoing challenges in India.
- Continuous glucose monitors produce ~288 readings a day — too many to review by hand.
- Question for the clinician: “Which monitored patient’s glucose pattern merits review, and what observations support that?”
- Our target: forecast glucose 30 and 60 minutes ahead and warn of a NEW sustained episode > 180 mg/dL (≥ 15 min), for Type 2 diabetes.

Use case and boundaries

- User: a clinician reviewing a small monitored cohort; secondary user: a reviewer inspecting evidence.
- Workflow (proposed): baseline labs at a visit → CGM + wearable measurements → twin forecasts → review list.
- Shows: observed vs predicted glucose, uncertainty, data freshness, factual reasons, research warnings.
- Never: insulin or medication advice, triage, live device claims, LLM-generated predictions.
- Potential fit with diagnostics + outpatient services is a proposal — no partnership or pilot is claimed.

What makes it a twin (operational definition)

- Versioned patient state: baseline profile, episode state, adaptation state, issued forecasts.
- Incoming observations released by a replay clock (only what was available at that time).
- Population forecaster + patient-specific correction learned from that person’s matured forecast errors.
- Measured updates over time — a statistical research twin, not a physiological simulator.

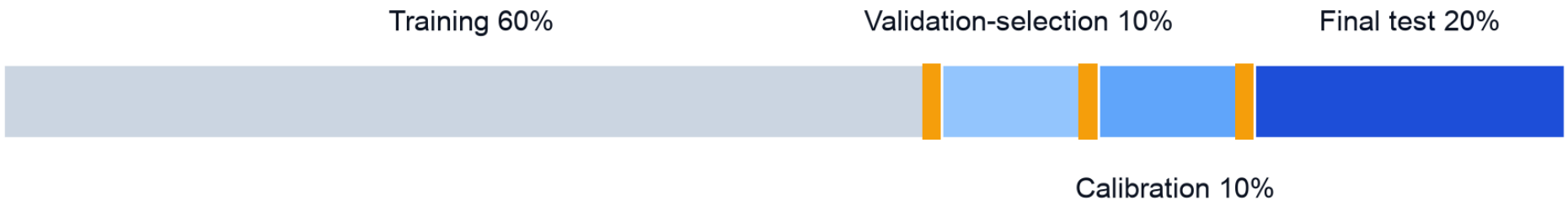


Data and provenance

- CGMacros 1.0.0 (PhysioNet, CC BY-NC-SA 4.0): 45 adults, ~10 days, Dexcom G6 Pro + Libre Pro (blinded), Fitbit, meals, screening labs.
- Primary cohort: 14 participants in the dataset's Type 2 group (HbA1c > 6.4 %); 31 others replayable but excluded.
- Finding: the published 1-minute CGM columns are linearly interpolated. We recover native 5-minute Dexcom readings (integer knots) and never count interpolated rows.
- Undocumented columns reported, not interpreted; documented error codes excluded; dates shifted; timezone unknown → synthetic clock, no time-of-day features.
- Synthetic fictional participants exercise edge cases (gaps, late meal logs, short history) with no download.

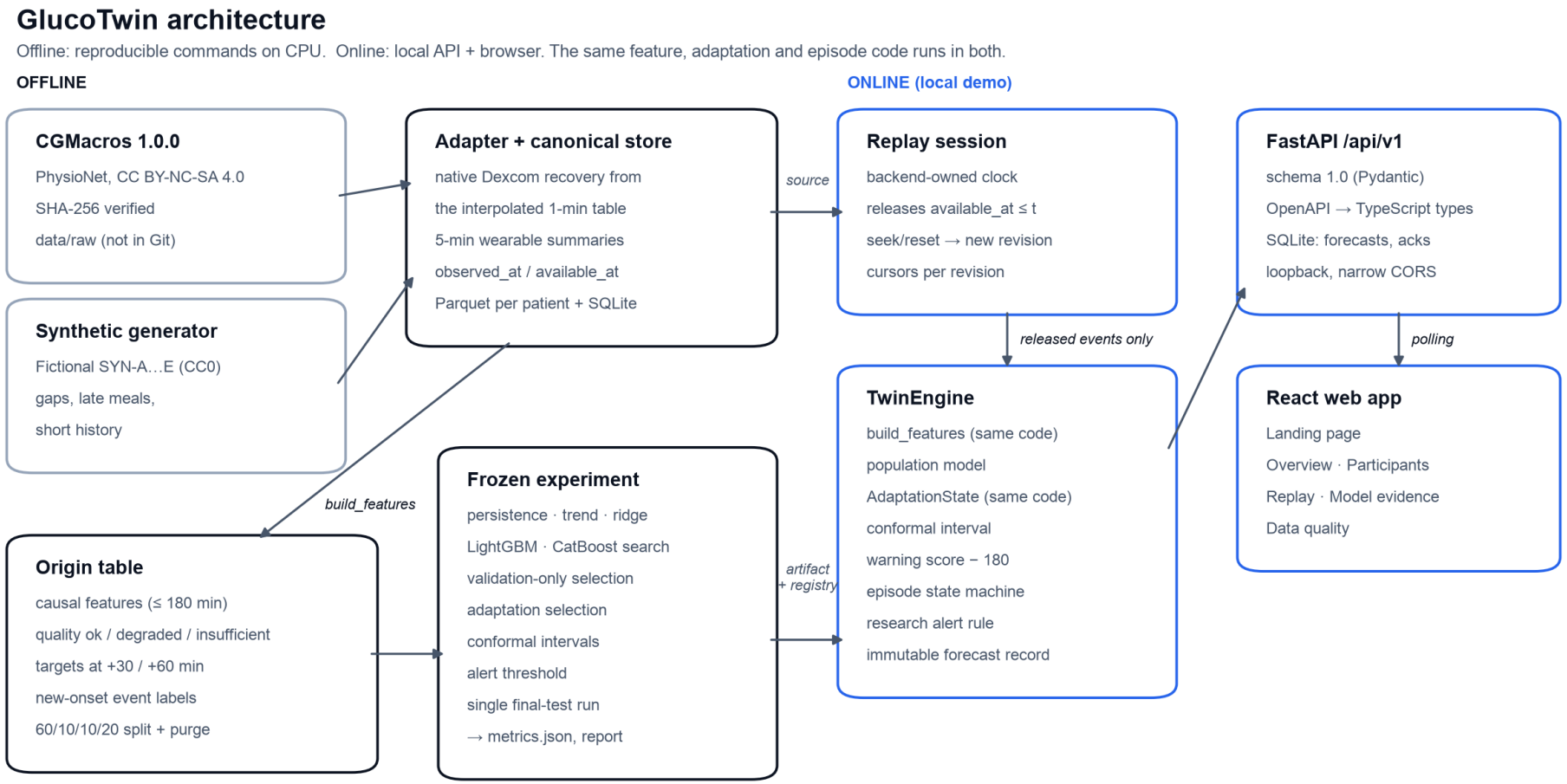
Information availability and leakage controls

- Every record carries observed_at and available_at; a forecast at t uses only records with both ≤ t.
- Meals excluded from the primary model (logging time unknown); tested separately as an optimistic sensitivity.
- Protocol frozen before fitting; model, adaptation, intervals and alert threshold chosen on validation. Results shown are a corrected re-run (gt-primary-2026-10-11-v2) of gt-primary-2026-10-11-v1 after two code fixes — same protocol, same test segment, already inspected.
- Adaptation uses a residual only after its target reading was available; missing future data is censored, never negative.



One participant's timeline (~10 days); every participant is split the same way. Amber: purged origins whose 60-min target + 15-min confirmation would cross a boundary.

Architecture



Models and personalization

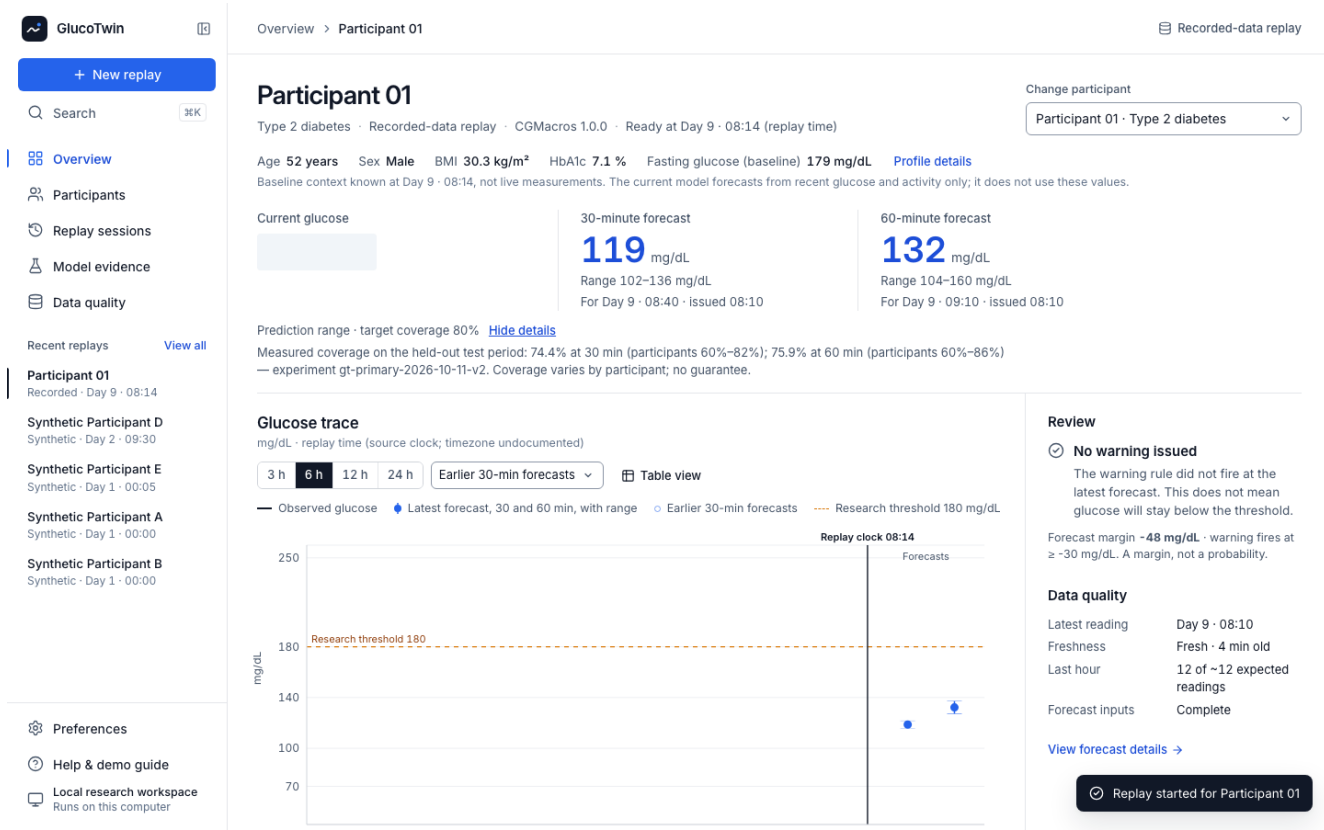
- Baselines: persistence, bounded linear trend, ridge regression.
- Direct per-horizon gradient-boosted trees: LightGBM (starting point) and CatBoost (challenger), equal 12-trial seeded searches, identical inputs and origins.
- Patient adaptation (same policy for both families): shrunk, bounded, decaying mean of matured residuals vs the frozen population forecast; weights never refit.
- Prediction ranges: split-conformal, target coverage 80 %, calibrated on the calibration segment for the exact serving pipeline.
- Warning: max forecast within 60 min – 180 mg/dL — an uncalibrated score, never shown as a probability.

Why this learner: validation-only selection

Candidate	Val. MAE 30	Val. MAE 60	Score
lightgbm:dyn-prof-v2	13.47	21.26	17.36
catboost:dyn-prof-v2	13.56	21.31	17.44
lightgbm:dyn-v2	**13.76**	**21.73**	**17.75**
catboost:dyn-v2	13.88	21.66	17.77
ridge:dyn-v2	14.13	22.44	18.28
ridge:dyn-prof-v2	14.21	22.61	18.41
linear_trend:cgm-only-v1	15.66	26.33	20.99
persistence:cgm-only-v1	16.51	26.32	21.42

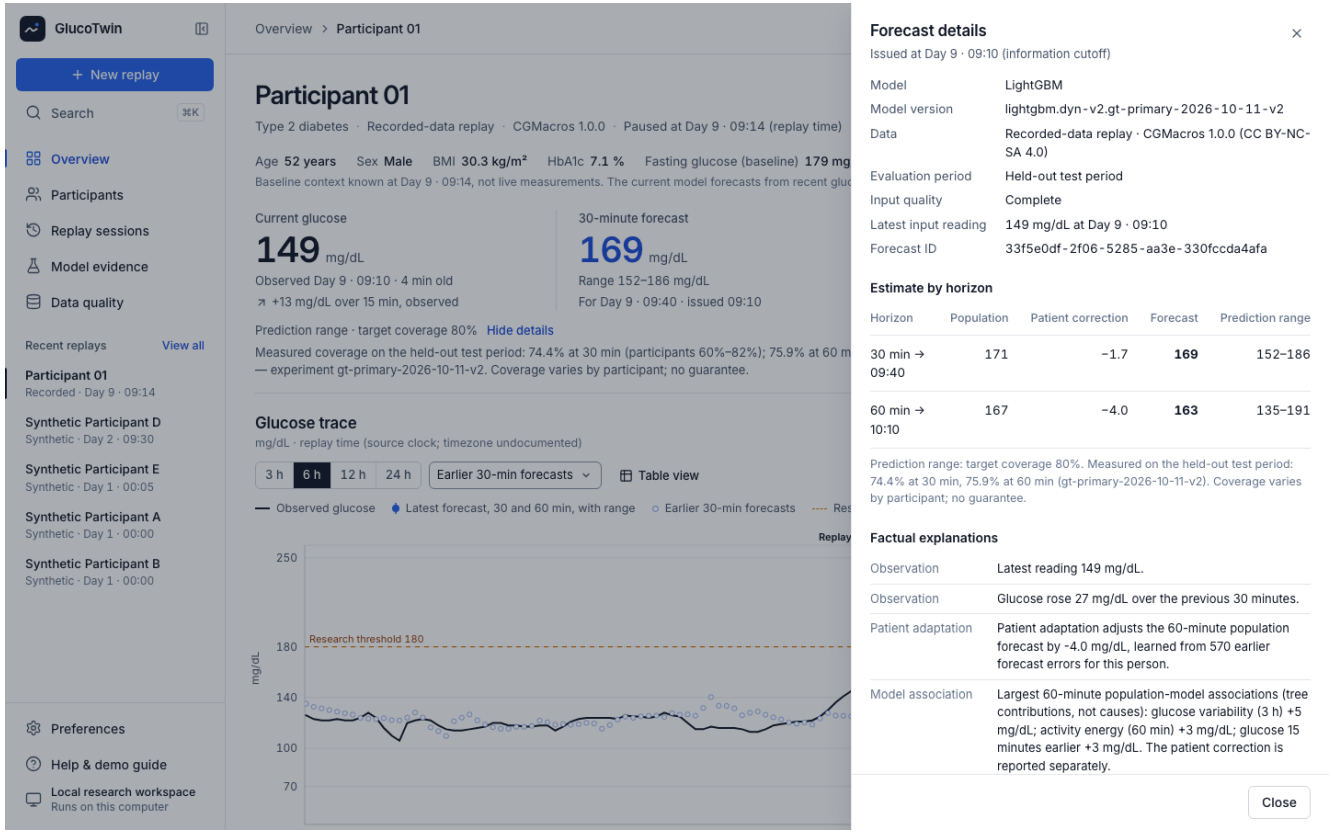
Declared rule: lowest mean per-participant MAE over both horizons; candidates within 0.5 mg/dL tie → simpler family, then smaller feature set. Selected: lightgbm:dyn-v2 with patient adaptation (adopted on validation). Best validation score: lightgbm:dyn-prof-v2 (17.36). The declared tie-break chose the simpler configuration; we did not switch after seeing test results.

Clinician workflow: the twin workspace



Recorded-data replay in the held-out test period: current glucose, 30/60-minute forecasts with prediction ranges (target coverage 80 %), saved earlier forecasts, research warnings with outcomes, review status and data quality.

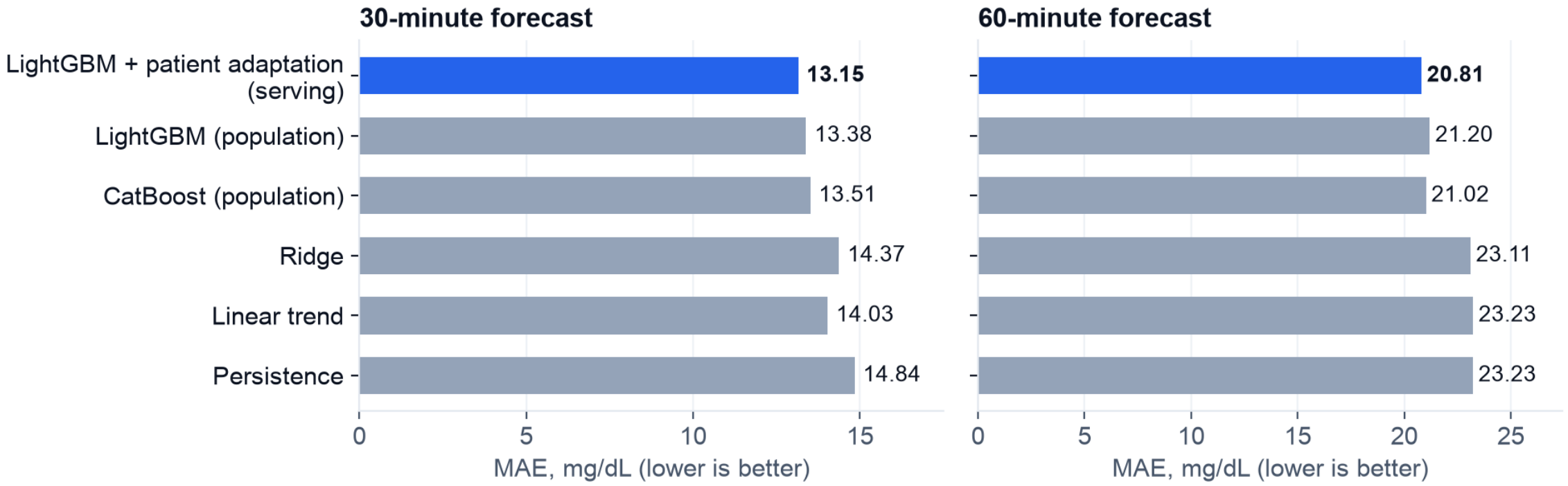
Demonstration: evidence behind a forecast



Each forecast is immutable and inspectable: information cutoff, model version, evaluation segment, population forecast vs patient correction, factual explanations.

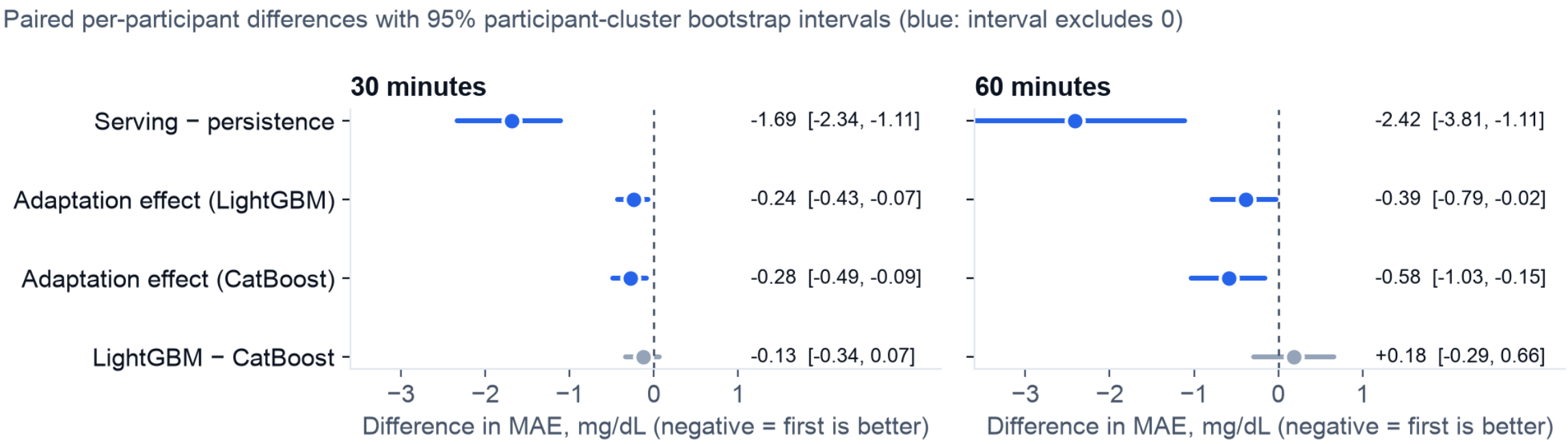
Measured evaluation: forecast accuracy (held-out test)

Held-out test: macro mean of per-participant MAE (14 participants)



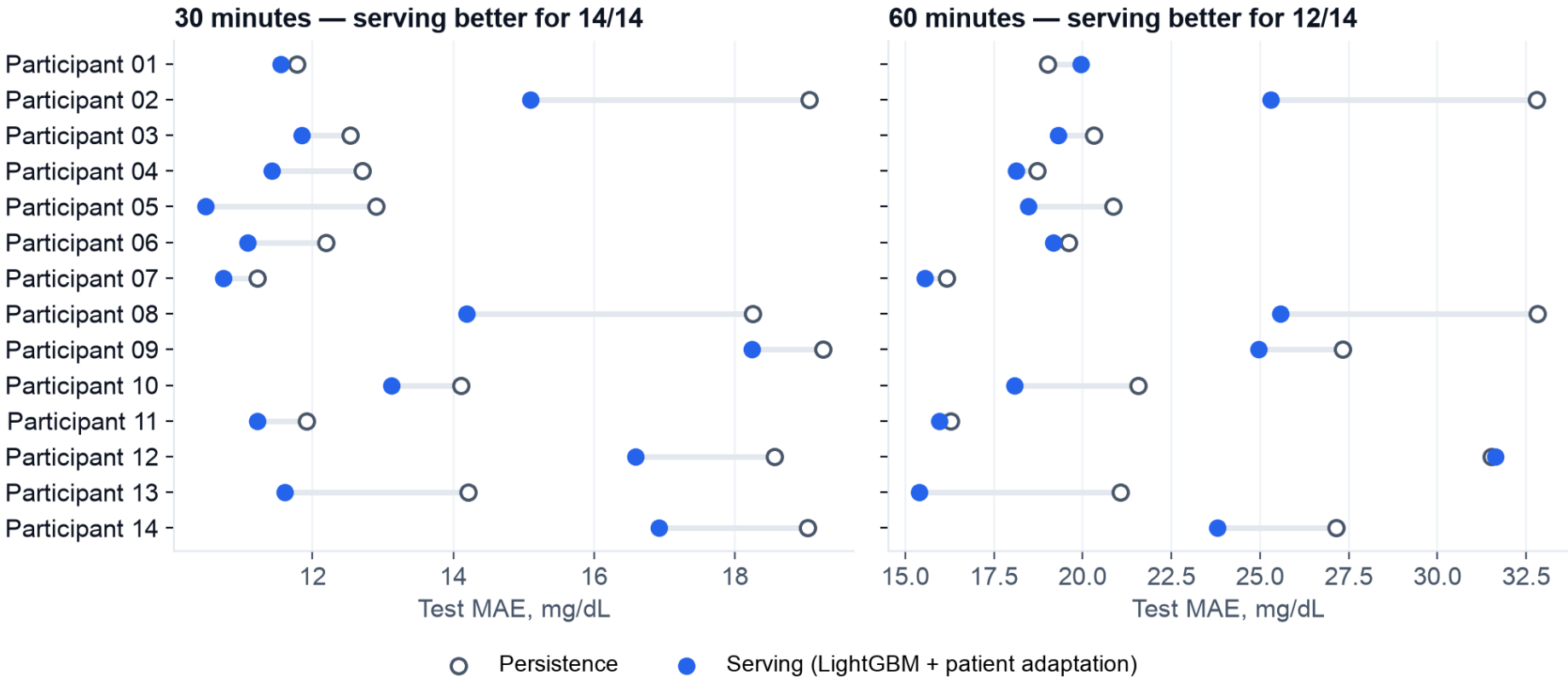
Serving – persistence: -1.69 mg/dL [-2.34, -1.11] at 30 min, -2.42 mg/dL [-3.81, -1.11] at 60 min. Test origins 7,730 (≈ 644 h); forecasts issued for 99.9 % of origins with a target.

Measured evaluation: does personalization help?



Adaptation (LightGBM): -0.24 mg/dL [-0.43, -0.07] at 30 min, -0.39 mg/dL [-0.79, -0.02] at 60 min — small; population 13.38 / 21.20, frozen state 13.40 / 21.16, adaptive 13.15 / 20.81 mg/dL. New-participant cold start: 14.92/25.47 → 14.07/23.12 mg/dL with adaptation.

Measured evaluation: every participant



Per-participant test MAE, persistence vs the serving pipeline. Gains are uneven: large for some participants, none for a few at 60 minutes. Fourteen people cannot establish population-level performance.

Measured evaluation: early warning and uncertainty

- 60-min warning (test, margin threshold -30 mg/dL chosen on validation by F1): 64/64 new onsets detected, precision 0.45, 79 false alarms (4.51 per eligible patient-day); forecast lead median 55 min (episode start 68 min).
- Same procedure on persistence (fires at glucose ≥ 150): 63/64 detected, 70 false alarms (4.00 per eligible day).
- Prediction ranges (target coverage 80 %) covered 74.4 % (30 min) and 75.9 % (60 min) on test — under-covered; no guarantee claimed.
- Optimistic meal logging: 18.87 vs 21.20 mg/dL at 60 min — timely meal data would likely help most.

New people and an independent cohort (research experiment)

- Three models, kept separate: LightGBM v2 is the demo model (all results above); a Ridge model on CGM-only features is the five-minute candidate; a separate 15-minute LightGBM is the external check.
- 145 participants imported; 14 evaluated internally (five participant folds, each person held out once) and 98 evaluated externally (ShanghaiT2DM, China, opened once).
- Held-out people, 30 / 60 min: candidate 13.88 / 22.30 vs demo recipe refit 14.52 / 24.25 vs persistence 15.34 / 24.47 mg/dL — but worse on large highs (+2.71 [+0.28, +5.11] mg/dL at 60 min when the target exceeded 180).
- Warnings chosen for ≤ 2 false alarms per day: 301/329 onsets at 1.31 false alarms per observed day (demo recipe: 329/329 at 2.80).
- External 15-minute LightGBM vs persistence: -0.99 [-1.21, -0.77] mg/dL at 30 min (clear); -0.34 [-0.76, +0.10] mg/dL at 60 min (inconclusive).
- Not promoted. CGMacros was already inspected, so its folds are internal validation; the next protocol also bounds high-glucose error, missed onsets and false warnings (declared, not run).

Project outcomes

- Built: a working personalized glucose twin — replay API with a backend-owned clock, immutable forecasts, research warnings with outcomes — and a clinician workspace (landing page, Overview, Participants, Replay sessions, Model evidence, Data quality).
- Found and corrected a data-provenance problem: the published 1-minute CGM values are interpolated; only native 5-minute readings are used.
- Measured on held-out test (corrected re-run): 14/14 participants improved over persistence at 30 min and 12/14 at 60 min; MAE 13.15 / 20.81 mg/dL vs 14.84 / 23.23.
- Personalization: small gains (-0.24 mg/dL and -0.39 mg/dL). Warnings: 64/64 onsets detected with 79 false alarms — the F1 rule favours sensitivity over review workload.
- Corrected after an independent review: an activity-window feature defect and an online warning double-credit were fixed and the protocol re-run (gt-primary-2026-10-11-v2); the original gt-primary-2026-10-11-v1 results are preserved.
- Reproducible: synthetic mode needs no download; recorded-data results regenerate from versioned configs; a fresh copy passes the automated backend, frontend and browser tests.

Limitations

- 14 participants from one US site; overlapping windows are not independent patients.
- No Indian participants — nothing here establishes performance for patients in India.
- Blinded research CGMs replayed with assumed zero latency; native timestamps known to the minute.
- Medication and insulin use unknown; meal logging time undocumented.
- Prediction ranges under-cover; the warning score is uncalibrated and its threshold rule (F1) ignores review workload; 180 mg/dL is a research threshold, not triage.
- Retrospective replay only: no clinician workflow, outcome or safety study. Optional Chronos-Bolt not run.

Next steps: evidence needed before use with Indian patients

- Prospective, consented Indian cohort with unblinded CGMs, recorded transmission latency, medications and meal logging times.
- Re-evaluate held-out participants (cold start) and per-site calibration; report subgroup performance.
- AI-READI five-minute CGM and wearable data (access not yet requested; steps listed) for a larger new-person evaluation under the declared protocol gt-nested-2026-10-12-v2.
- Clinician usability study of the review workflow; regulatory and data-protection review (DPDP Act).
- Optional: compare a fine-tuned time-series foundation model (e.g. Chronos-Bolt) under the same protocol.

Team, links, reproducibility and licenses

- Team Kryptonite — Avinash Gehi (team leader) · Hardik Hinduja. College / incubator: to be confirmed by the team.
- Code: public GitHub link — added after the team publishes the repository — MIT license (code only). Quick start: make setup · make synthetic · make dev.
- Demo video (20:00): unlisted YouTube link — added after recording and upload. Architecture: docs/submission/GlucoTwin_Architecture.pdf.
- Data: CGMacros 1.0.0, Gutierrez-Osuna, Kerr, Mortazavi, Das (2025), PhysioNet, doi:10.13026/3z8q-x658, CC BY-NC-SA 4.0; ShanghaiT2DM, Zhao, Zhu, Wang, Rao (2022), figshare, doi:10.6084/m9.figshare.21600933.v5, CC BY 4.0.
- All results: experiment gt-primary-2026-10-11-v2, artifacts/experiments/…/metrics.json; reports/evaluation_report.md.